

Experiment 11 — MongoDB Databases, Collections and Records

GitHub-readable preview of the presentation. Download the matching .pptx for the editable PowerPoint.

1. Syllabus tasks

- Create a database context.
- Create a students collection.
- Insert multiple records.
- Find records.
- Limit the number of results.
- Sort by marks.
- Create an index.
- Calculate average marks with aggregation.
- Average marks for each branch.
- Number of students in each branch.

2. Workflow

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Workflow: input → processing → data/state → output

3.1. Code pathway — queries.js

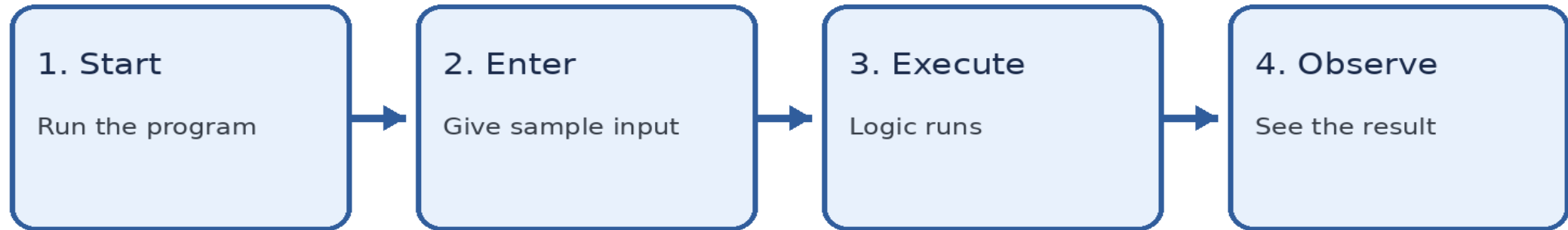
```
1 /**
2  * Experiment 11: MongoDB Databases, Collections and Records
3  *
4  * Run this file inside mongosh.
5  *
6  * Demonstrates:
7  * - Database and collection creation
8  * - Insert many records
9  * - Find
10 * - Limit
```

3.2. Code pathway — queries.js

```
11 * - Sort
12 * - Index
13 * - Aggregation
14 */
15
16 // -----
17 // Database and collection
18 // -----
19
20 use fsd2records;
```

4. Runtime walkthrough

Runtime walkthrough — Experiment 11 — MongoDB Databases, Collections



Runtime: start → enter → execute → observe

5. Run and observe

- Start the server / Vite app or open the MongoDB shell as described in the experiment README.
- Enter the sample input or call the required route.
- Watch the terminal for logs and errors.
- Verify the browser, API response, or database result.

6. Viva questions

- What is the purpose of this experiment?
- Which input is required?
- Which line or command performs the main operation?
- What output should you observe?
- What happens when the input changes?